

Reflection Journal

Course / Lab: ITAI 1371 — Module 3: Machine Learning Workflow and Types of Learning

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Introduction

This lab helped me move from learning individual machine learning concepts to understanding how the different steps work together as a complete workflow. At first, the machine learning process seemed like a number of separate technical steps. As I worked through the notebook, I began to see how defining the problem, examining the data, selecting features, training a model, evaluating performance, and reflecting on the results are connected.

Understanding the Types of Learning

One of the most important concepts I learned was the difference between supervised, unsupervised, and reinforcement learning. Supervised learning uses labeled data to learn how to make predictions or classifications. Unsupervised learning looks for patterns or groups in data without labeled outcomes. Reinforcement learning is based on an agent interacting with an environment and learning from rewards or penalties. The distinction became clearer when I thought about whether the training data provides an answer or target for the model.

What I Learned from the Wine Classification Project

The wine classification project gave me an opportunity to see supervised learning in practice. The dataset contained 178 observations, 13 features, and three wine classes, with no missing values. I examined the data and visualizations before comparing Logistic Regression and Decision Tree models. Logistic Regression performed better in this lab, with an accuracy of 88.9%, compared with 83.3% for the Decision Tree. This showed me that different algorithms can perform differently on the same dataset.

Feature Selection and Model Performance

I also learned that the features included in a model can affect its performance. I experimented with alcohol, color intensity, and proline as selected features. The resulting model accuracy was 83.3%, lower than the original 88.9% accuracy. This helped me understand that reducing the number of features does not automatically improve a model. Some of the other available features may contain useful information for distinguishing between the wine classes.

Challenges and Problem-Solving

The most challenging part of the lab was understanding how to choose features and compare model performance. I also had to become more comfortable interpreting accuracy rather than simply running the code. Working through the notebook step by step helped me overcome this challenge. I learned to look at the code, understand what the model was doing, and use the results to guide the next step.

Connection to Banking and Finance

I can connect these concepts to banking and finance. A practical machine learning problem would be detecting fraudulent financial transactions and identifying unusual patterns in customer transactions. This would be a supervised classification problem when historical transactions are labeled as fraudulent or legitimate. Useful data

could include transaction amounts, dates and times, locations, payment methods, and information about previous fraudulent transactions.

Questions for Further Investigation

Two questions I would investigate further are: (1) How would using all 13 wine features affect the accuracy of the Logistic Regression model compared with using only three selected features? (2) How would other machine learning algorithms, such as Random Forest or Support Vector Machine, perform compared with Logistic Regression on the wine dataset? These questions would help me understand how feature selection and algorithm choice influence model performance.

Conclusion

Overall, this lab changed my view of machine learning from a collection of coding techniques into a structured problem-solving process. I learned that understanding the data and the problem is just as important as selecting an algorithm. I also learned that model performance should be evaluated and compared rather than assumed. Most importantly, I now have a clearer understanding of how supervised, unsupervised, and reinforcement learning differ and how machine learning can be connected to real-world applications in banking and finance.